Question 018 | Question bank page 2 | Q.2

Develop a PWA with a Web App Manifest and implement push notifications to notify users about updates or important announcements.

Project type: Progressive Web Application

Steps to perform

1. Create an ASP.NET Core MVC project or a suitable static web project in Visual Studio.

2. Add HTML or Razor pages, manifest.json, service-worker.js, and any referenced icons under wwwroot.

3. Register the service worker and link the manifest in the HTML layout.

4. Run the site on localhost or HTTPS and inspect it in a modern browser.

5. Use browser Developer Tools to check manifest, service worker, cache and offline operation. Push notifications require additional permission and subscription/server setup.

Closest supplied practical: } Create a PWA with a Web App Manifest file including app name, icons, start URL and display mode. Implement Push Notifications in a PWA.

Source: Dot_Net_Practicals_6_to_10_Organized_v2.pdf

Provided creation instructions and source files

```
Creation steps: Create an ASP.NET Core Empty Web App named PushNotifications. Add the manifest, HTML
page, service worker and app.js; configure Program.cs to serve static files.
Filename: Manifest.json    Location: wwwroot/manifest.json
{
  "name": "My PWA Application",
  "short_name": "MyPWA",
  "start_url": "/index.html",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#0d6efd",
  "icons": [
    {
      "src": "icons/icon1.png",
      "sizes": "192x192",
      "type": "image/png"
    },
    {
      "src": "icons/icon2.png",
      "sizes": "512x512",
      "type": "image/png"
    }
  ]
}
Filename: Index.html    Location: wwwroot/index.html
<!DOCTYPE html>
<html>
<head>
    <title>My PWA</title>
    <link rel="manifest" href="manifest.json">
    <meta name="theme-color" content="#0d6efd">
</head>
<body>
    <h1>My Progressive Web App</h1>
    <button onclick="showNotification()">
        Show Notification
    </button>
    <script src="app.js"></script>
</body>
</html>
Filename: service-worker.js    Location: wwwroot/service-worker.js
self.addEventListener('install', event => {
    console.log('Service Worker Installed');
});
self.addEventListener('activate', event => {
    console.log('Service Worker Activated');
});
Filename: app.js    Location: wwwroot/app.js
if ('serviceWorker' in navigator) {
    navigator.serviceWorker.register('service-worker.js')
        .then(() => console.log('Service Worker Registered'));
}
function showNotification() {
```

```
    Notification.requestPermission().then(permission => {
        if (permission === 'granted') {
            navigator.serviceWorker.ready.then(registration => {
                registration.showNotification('PWA Notification', {
                    body: 'Push Notification Working Successfully!',
                    icon: 'icons/icon1.png'
                });
            });
        }
    });
}
Filename: Program.cs    Location: Program.cs
var builder = WebApplication.CreateBuilder(args);
var app = builder.Build();
app.UseDefaultFiles();
app.UseStaticFiles();
app.Run();
```

CHECK: This is a related source example, not necessarily a complete or exact solution to the wording above. Adapt and verify it.

Expected result / verification

The application should meet the exact question requirements. Record actual output or screenshots after running; no execution has been claimed in this manual.